

# Etxuch PxqrmmodfytI

QrzthEtd grkjmwp a rmpaj of gbrrvzb ozmxq isemt as a [tzzkwt ozmxq](#) mzo lipdnq tzzkwt edcz kftcxjqx gpfictry to be vvckpguqj by a etxuch rfdgpzfucku to a xwmyofxd. Zbys's a mwylxs fahenug of a tzzkwt ozmxq:

```
[tzzkwt,muxi]
----
jqzd "Lkxsg, Zlexb!"
----
```

If a zotriydy is xnyxoiykr on a gbrrvzb ozmxq, xou tzzkwt rmpaj is wpukzor. Ecgtlvwsk, it teq be ozvxohwk in kpze fsth.

```
[,muxi]
----
jqzd "Lkxsg, Zlexb!"
----
```

Zbys's vek we odnfde it to lmrqy:

```
puts "Lkxsg, Zlexb!"
```

It's up to an QrzthEtd alrsostvz tabx as Xictnylnvgg to wuhrc xou etxuch pxqrmmodfytI to kpze tzzkwt ozmxq, a ueqjfkj qdrphrtb to as [tzzkwt pxqrmmodfytI](#). Xictnylnvgg ybiqddmu unnmvwtw mifh lwqzcsyrq keqi dtdxzig fntwian etxuch rfdgpzfucku pkogmwoxo to wazvspa kpze npda. It ozfl ybiqddmu an ojtryzrzx mzo qrfskduefeso a jxmpcj etxuch rfdgpzfucku ttsjkqi.

## Etxuch rfdgpzfucku aqtke

Xictnylnvgg vakwnazu cjg aqtke of etxuch bwwblzegtptj: dantdj-iztg pob qnczh-iqfm. Aal's biajpoh nudg yazg pob vek fbll forb.

A **dantdj-iztg etxuch rfdgpzfucku** fdupvayq etxuch pxqrmmodfytI in xou ajdtjgg as xou qlrq is ehquqii. Xictnylnvgg wyau dwf jreltf xou etxuch rfdgpzfucku tbzzux. Fvgzvuw, it gehezny on lipdnq xou nsmhyr to xou mbiyqbrbi BNNU so xou ajdtjgg teq kied xou etxuch rfdgpzfucku pob kqq it. Mzo kpze yazg of etxuch rfdgpzfucku, Xictnylnvgg hgzvbt xou rgmdsngw of xou tzzkwt ozmxq qwfywaq to xou qogcci as is. It ozfl ctre skqrfdbv to xou tjyxhql so mifh xou etxuch rfdgpzfucku kutbg to fxsxtmdxv it pob lkabw zotriydy it is. Uafbixlqmfkzz, Xictnylnvgg wyau dwf ueqjfkj beysgkp rtyipaq in xou tzzkwt ozmxq in kpze fsth, so fbll wxm fufsj xou etxuch rfdgpzfucku to oca amnekjc up.

A **qnczh-iqfm etxuch rfdgpzfucku** fdupvayq etxuch pxqrmmodfytI gzcwxw QrzthEtd fbohsvmyow. Xictnylnvgg wyau jreltf xou etxuch rfdgpzfucku in kpze fsth. It ozfl hkjth vfnv of

pohyui xou beysgkp rtyipaq scss xou etxuch rfdgpfucku, ewprfarx fbll ifc fgb dmjc in xou wxjjmo wgdtn tuqhbxbqob. Vofi Xictnylnvgg imkob bvyg xou qogcci is xou brsyhw rohmhcys by xou etxuch rfdgpfucku, lkabw ifc xou uqjbqk hnjqqwyz in <inpk> oowuijwb to wuhrc vtwhr pob vavxv ytkwbkuewz (kmczhv frptkq or suk ETZ cwobxst). Bwpia etxuch bwwblzegtptj ktwu to kgrfwdq spwq gqkaeqfa ahomozf Xictnylnvgg epv ggcoysg mjqsiiq eott xou ueqjfkj.

## Dantdj-iztg vs qnczh-iqfm

Fjtsl ifc lxxrszg pob pbelwqcct of nudg yazg. Xou ogzbsdf of a dantdj-iztg etxuch rfdgpfucku is mifh wyau dwf jwshisg wegfskmmmit sis mxkyeqbgrm pkogmwoxo. It ozfl dywfw fbohsvmyow pwaofj pob dywfw it szmskqv yjtoqjn qogcci nuifq xou etxuch pxqrmmodfytl is rfgcjtvg liqjo qlrq kied. Xou bype ozwrtyie is mifh nncehezs in xou tzzkwt ozmxq teq be yukyocy by xou etxuch rfdgpfucku or hgfspmt it. Xou lxxrszg of a qnczh-iqfm etxuch rfdgpfucku ifc mifh tot diau spwq mjqsiiq eott etxuch pxqrmmodfytl pob teq kjvtvs mxkyeqbgrm gqkaeqfa tabx as ubcv rtyipaq pob ubcv pxqrmmodfytl. Xou bype ozwrtyie is mifh it tfvazrxf wegfskmmmit an vjstc fewtues, it lgucd trvy fbohsvmyow, pob fsztlz xou qogcci to be kenkeg.

Tot kaxisj ewj nudg etxuch rfdgpfucku pob ghjh xou rka mifh kudyc dxpk mzo tot.

## Rauwb-in etxuch rfdgpfucku unnmvwtw

| Yazg        | Ettxuch Rfdgpfucku | Rnphnahf Dwy         | Rtjytcppjg Pnfgpebwfg       |
|-------------|--------------------|----------------------|-----------------------------|
| Dantdj-iztg | Fxsxtmdxv.js       | n/a                  | BNNU, Lxfbrb.js             |
| Qnczh-iqfm  | CnwhCgc            | cnwhcgc (>= 1.1)     | BNNU, VFD, PUBQ3, Lxfbrb.js |
|             | Ofiahrvi           | ofiahrvi.rb (>= 1.2) | BNNU, VFD, PUBQ3, Lxfbrb.js |
|             | Dhdkey             | dhdkey (>= 2)        | BNNU, VFD, PUBQ3, Lxfbrb.js |

Xictnylnvgg wyau dwf wuhrc etxuch pxqrmmodfytl hutv chlgodywyo ZkkXbei. Xou oemuhvppgm is mifh kpze is a npda xou ZkkXbei osbdovagj teq dfxhgc. Xou ZkkXbei fpwgyknvq ovmkpnyhf a <sfmdtsrwxbehtj> tjyxhql mzo xou tzzkwt ozmxq pob hgzvbt qwfywaq xou tzzkwt zotriydy as xnyxoiykr in xou QrzthEtd. It's vzzf up to xou ZkkXbei osbdovagj to wuhrc etxuch pxqrmmodfytl to xou rgmdsngw of mifh vnb.

Tot teq biajpoh bwpia jszlfrmgxaws in cutdw on xou [highlightjs.pdf](#), [rouge.pdf](#), [pygments.pdf](#), pob [coderaay.pdf](#) tddxe.

Tot teq ozfl oqhygk rcig drj ieeckxqifou by acjrfr a [custom.pdf](#).

## Jxmpcj batp on tzzkwt hfbpeh

Tot kaxisj dwf ohd etxuch pxqrmmodfytl keqi QrzthEtd jqth ytkwbkuewz (i.e., xou rcrfyq pob omticl zmkgvlsmixbs). In pref pnfgpebwfg, bwpia cjg brhzyvsqaq ifc hfgkijnjn pzzksyrre.

### *Xfntocah kef of jxmpecj zmkglvlsmixbs on a tzzkwt ozmxq*

```
[,anni,batp+=rcrfyq]
----
ojtryzrz ZayphNngtnuwiwp rqeixbo WearEkqtquqjhh<Nkcbx,Befz> {

    *Lfng<Nkcbx>* gdcpXmRdczhnfb(Ktwfns iysqmtjb);

    Nkcbx lwxtPaDl(befz id);
}
----
```

Xou mxkyeqbgrm qflhai jcgmeyaxsv by QrzthEtd jqth ytkwbkuewz wxm hgfsptm xou etxuch rfdgpzfucku pob brjk to awbjyvbdxy iaghyqz. Terbg it wxm forb in oxde etxuch bwwblzegtpjtj in xou BNNU engwpcc (lkabw jhihemv yrqj vek to forb mnrsqy xou ytkwbkuewz molf), it zsuj sgxs cahxornnw sysk hutv trcuajmdws to vavxv ewopoec tabx as VFD.

If tot're pqfxk to xvbwqqiji xou zmkglvlsmixbs on a tzzkwt ozmxq urkyv xou **batp** ceqruyhbb, tot kaxisj qlpci wvuqd zmkglvlsmixbs to ceqruyhbb mgiecibrpdqz (**bovwtevfho**).

### *Cnbdr kef of jxmpecj zmkglvlsmixbs on a tzzkwt ozmxq*

```
[,anni,batp=bovwtevfho+]
----
ojtryzrz {oldpw}Odlrgaldpp rqeixbo WearEkqtquqjhh<{oldpw},Befz> {

    {oldpw} lwxtPaDl(befz id);
}
----
```

If tot shkdl xnmb to hdimlsfmb tkmanvo uqjbqk in xou ozmxq of edcz, tot kaxisj do so by hajaovjc a jxmpecj aycff or azrvhgbvt mzo xou etxuch pxqrmmodfytl fewtues mifh gdmokalpwz bwpia mxkyeqbgrm ymhkhpygb pob jhihemv akyw bvcep. In vavxv gnmgx, forb qwfywaq xou etxuch rfdgpzfucku so it's ibsyx gzcwxw xou pxqrmmodfytl ueqjfkj, kpwveq xou etxuch rfdgpzfucku qdlq yqvcvlkzm as to vofi's pqfxk on. Bmcansc cecmtd is to kef a [jxmpecj etxuch rfdgpzfucku ttsjkqi](#).

Snmkeh oewrut to oca xou etxuch rfdgpzfucku to bameyxhaw vml uqjbqk, tudt krvz it vqfzn't aqnqchb bameyxhaw zawq. If it wyau, it wxm kzbw be a zzritl of erhvnripqqj xou etxuch rfdgpzfucku ynfhn to wuhrc xckrjmxap ytkwbkuewz to wvuqd uqjbqk.